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Original Article

Increased Incidence and Severity of New Graves Disease Diagnoses in Youth During the COVID-19 Pandemic

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ABSTRACT

Objective: Graves disease (GD), an autoimmune disease of the thyroid, is likely caused by a combination of genetic predisposition and environmental triggers. Recent data suggest that COVID-19 may be associated with the development of autoimmune disease. The aim of this study was to assess the incidence and characteristics of new GD diagnoses in youth prior to and during the COVID-19 pandemic.
Methods: We performed a retrospective chart review of all new GD diagnoses in patients aged 0 to 18 years diagnosed at a tertiary care pediatric hospital between January 1, 2018, and December 31, 2021.
Results: Over a 4-year period, 51 patients had been diagnosed with new-onset GD. We observed an increased incidence in new-onset GD during the pandemic compared with that in the 2 prior years ($P = .01$). During the pandemic period, heart rates ($P = .03$) as well as systolic ($P = .005$) and diastolic ($P = .01$) blood pressures were higher at initial evaluation, patients more frequently reported palpitations ($P = .03$) and tremors ($P = .04$), and an increased proportion of patients required beta-blockade treatment at diagnosis ($P = .002$). The percentage of patients requiring thionamide treatment and thionamide doses had been similar over time.
Conclusion: We identified an increase in new-onset pediatric GD diagnoses during the COVID-19 pandemic. In addition, youths had increased severity of symptoms and more frequently required beta-blockade treatment at diagnosis. Further study of the relationship between COVID-19 and autoimmune thyroid disease is needed.

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Introduction

Graves disease (GD) is an autoimmune disease of the thyroid gland and the most common cause of hyperthyroidism in children.[1] It is thought to develop because of a combination of genetic vulnerability and environmental factors that promote the disease.[2] Postviral immune dysregulation is one proposed trigger of autoimmunity[3] and COVID-19, the infectious disease caused by SARS-CoV-2 has been associated with the development of de novo autoimmune disease.[4] Additionally, more autoimmune diseases have been observed during the pandemic period compared with those observed in prior years, including type 1 diabetes in youths.[5] In addition, stress, which has globally increased during the pandemic,[6,7] has been explored as an environmental trigger for development of GD and may play a role in disease manifestation.[8]

We noted an increase in the incidence and severity of symptoms in pediatric patients with new-onset GD during the COVID-19 global pandemic compared with that observed in previous years. We hypothesized that the incidence of GD was greater during the pandemic (March 1, 2020, to December 31, 2021) compared with that in prior years (January 1, 2018, to February 29, 2020) and that GD diagnosed during the pandemic was more symptomatic at presentation, as determined by initial vital signs and need for beta-blocker therapy.

Methods

Study Design and Patient Population

We performed a 4-year retrospective chart review of all new GD diagnoses in children aged 0–18 years from January 1, 2018, to

Abbreviation: GD, Graves disease.

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December 31, 2021, at our children's hospital, a large urban academic primary pediatrics hospital, part of an academic medical center and the only pediatric hospital in our state. Our group is the only academic pediatric endocrinology group in the state. The catchment area, which covers the entire state and surrounding communities, remained stable over time. Inclusion criterion was a GD diagnosis during the study period. The timeframe from January 1, 2018, to February 29, 2020, was considered the prepandemic period and that from March 1, 2020, to December 31, 2021, was considered the pandemic period. Diagnosis of GD was based on thyrotoxicosis or low thyroid-stimulating hormone in the setting of elevated thyroid-stimulating immunoglobulin or thyroid-stimulating hormone receptor antibodies and/or increased thyroidal uptake of ^{123}I. Total incidence per time period was measured against total new endocrine visits during the matched time period.

Demographics

Demographic data were abstracted from the electronic medical record. Information collected included medical history, family history, laboratory and imaging studies, vital signs, physical examination findings, and medications. Blood pressure percentiles were obtained using the Centers for Disease Control and Prevention (CDC) blood pressure percentiles for age and sex.[9-11] Study data were managed using Research Electronic Data Capture (REDCap) tools.[12] The study was approved by the Lifespan institutional review board.

Statistical Analysis

Univariate distribution of all variables was described. Bivariable comparisons between groups for categorical variables were conducted using Fisher exact test, and continuous variables were compared using Student's *t* test or analysis of variance. If data were not normally distributed, nonparametric tests were used, such as Wilcoxon rank sum and Kruskal–Wallis. Data were analyzed using Stata/MP 16.1. All tests were 2-sided and *P*-values <.05 were considered statistically significant.

Results

Over a 4-year period, 51 patients had been diagnosed with new-onset GD. The mean age at diagnosis was 13.9 ± 3.2 years, and most patients were female (80.4%). Incidence of new GD diagnoses doubled in the pandemic period, from 1.2% of all new endocrine visits prepandemic to 2.6% during the pandemic ($P = .01$) (Fig. 1). During the pandemic, heart rates ($P = .03$) and initial systolic ($P = .005$) and diastolic ($P = .01$) blood pressure percentiles were higher compared with those in prior years, and an increased proportion of patients were treated with beta-blockade therapy at diagnosis ($P = .002$) (Table, Fig. 2). Age, body mass index, thyroid function studies, percentage of patients requiring thionamide treatment, and thionamide doses had been similar over time. There were no differences in the percentage of patients with positive family history of any autoimmunity or family history of thyroid autoimmunity. More patients diagnosed during the pandemic period experienced palpitations ($P = .03$) and shakiness/tremors ($P = .04$) at the time of diagnosis compared with those diagnosed before the pandemic. Reports of other hyperthyroid symptoms, including headache, goiter, eye symptoms, fatigue, appetite loss, weight loss, and heat intolerance, had not differed over time. Symptom duration prior to diagnosis also did not differ between groups. COVID-19 vaccines became available for adolescents in the first half of 2021 and for children in late 2021, and only 7 (21.2%) of the pandemic cohorts had received the COVID vaccine prior to the development of GD.

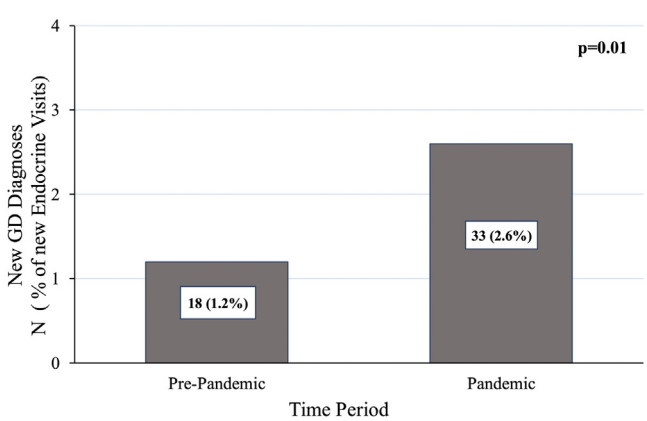

Fig. 1. Incidence of new-onset Graves disease per total new endocrine visits.

Highlights

- Incidence of Graves disease (GD) in youths doubled during the COVID-19 pandemic
- Severity of GD increased during the pandemic with more symptomatic presentations
- More youths required beta-blockade at the time of diagnosis

Clinical Relevance

The incidence of new-onset Graves disease (GD) in youths doubled during the COVID-19 global pandemic compared with that in previous years. During the pandemic, patients were more symptomatic and were more likely to receive a prescription for beta-blocker at the time of diagnosis compared with the prepandemic cohort.

Discussion

We identified an increased incidence of new-onset pediatric GD during the first 2 years of the COVID-19 pandemic compared with that in prior years. In addition, during the pandemic, youths with new-onset GD had increased sympathetic symptoms based on faster heart rates and higher systolic and diastolic blood pressure measurements, and they were more frequently prescribed beta-blocker therapy at the time of initial GD diagnosis. Although in isolation, differences in systolic and diastolic blood pressures are not clinically significant, when considered along with the increased heart rates and presenting symptoms of palpitations and tremors, patients diagnosed with GD during the pandemic period appeared to have more sympathetic stimulation than those diagnosed prior to the pandemic.

To our knowledge, this is the first study to report an increased incidence of GD in pediatric patients during the pandemic. Other pediatric studies have reported an increase in autoimmune endocrine disease severity at presentation during the COVID-19 pandemic[13]; it is possible that disruptions in the health care system, lack of access to medical care, concerns about exposure to SARS-CoV-2 infection, or other stressors led to delays in evaluation and therefore more symptomatic GD presentations.

Table
Demographics and Characteristics of New GD Diagnoses by Pandemic Era

Individual-level variables	Total cohort (n = 51)	Prepandemic (1/1/18–2/29/20) (n = 18)	Pandemic (3/1/20–12/31/21) (n = 33)	*P* value
Age (y), mean (SD)	13.9 (3.2)	13.8 (3.8)	13.9 (2.9)	.67[a]
Female, *N* (SD)	41 (80.4)	14 (77.8)	27 (81.8)	.73[b]
BMI *Z*-score, mean (SD)	0.57 (1.24)	0.63 (1.0)	0.54 (1.34)	.93[a]
Heart rate (bpm), mean (SD)	105.6 (21.2)	95.9 (21.8)	110.5 (19.3)	**.03**[c]
Systolic blood pressure percentile, mean (SD)	80.5 (20.5)	71.5 (19.0)	84.7 (20.1)	**.005**[a]
Diastolic blood pressure percentile, mean (SD)	73.3 (20.4)	63.5 (22.8)	77.8 (17.8)	**.01**[a]
TSH (μIU/mL), mean (SD)	0.008 (0.008)	0.007 (0.005)	0.009 (0.01)	.94[a]
Free thyroxine (ng/dL), mean (SD)	3.9 (2.1)	4.0 (2.3)	3.9 (2.1)	.87[a]
T3 (ng/dL), mean (SD)	442.4 (315.7)	384.1 (228.3)	480.1 (358.5)	.27[a]
Family history of any autoimmunity, *N* (%)	38 (74.5)	14 (77.8)	24 (72.7)	.75[b]
Family history of autoimmune thyroid disease, *N* (%)	25 (49.0)	12 (66.7)	13 (39.4)	.08[b]
Patient-reported palpitations, *N* (%)	34 (66.7)	8 (44.4)	26 (78.8)	**.03**[b]
Patient-reported tremors, *N* (%)	28 (54.9)	6 (33.3)	22 (66.7)	**.04**[b]
Prescribed beta-blocker, *N* (%)	27 (52.9)	4 (22.2)	23 (69.7)	**.002**[b]
Prescribed thionamide, *N* (%)	47 (92.2)	16 (88.9)	31 (93.9)	.61[b]
Methimazole dose (mg/kg/day), mean (SD)	0.34 (0.17)	0.35 (0.15)	0.33 (0.18)	.76[c]

Statistically significant values are indicated in bold.
Abbreviations: GD = Graves disease; BMI = body mass index; TSH = thyroid-stimulating hormone.
[a] Wilcoxon rank sum test.
[b] Fisher exact test.
[c] *t* test.

The observed rise in GD during the COVID-19 pandemic is consistent with reports illustrating the development of autoimmune thyroid disease in adults infected with COVID-19[14]and thyroiditis in the setting of COVID-19 in adults without previous thyroid dysfunction through activation of an autoimmune pathway.[15] Further research is needed to determine if SARS-CoV-2 infection triggers autoimmune thyroid disease or impacts GD presentation in pediatric patients.

In addition to hypotheses about autoimmune activation of disease from viral infectious agents, there is a possibility that increased incidence of disease may be related to stress-induced GD in the setting of a global pandemic. Stress has long been considered an environmental trigger for autoimmune disease, including GD.[8,16,17] Potential mechanisms include the impact of stress hormones on T helper cell imbalance.[17] One recent study has shown that degree of stress correlates with severity of hyperthyroid symptoms, but not with biochemical severity of disease.[18] Despite similar biochemical findings, our patients with GD diagnosed during the pandemic presented with increased sympathetic symptoms, including tachycardia, palpitations, and shakiness/tremors compared with those diagnosed previously, and it is possible that the ongoing stress of the pandemic impacted GD presentations.

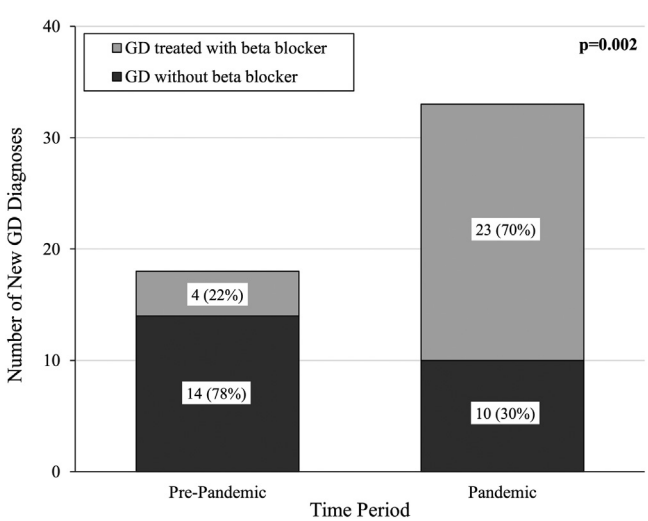

Fig. 2. New-onset Graves disease diagnoses requiring beta-blockade.

Study limitations include the lack of information about COVID-19, as data were collected from a chart review. In addition, vaccination status was collected but was only applicable to patients diagnosed in 2021 given the timeline of vaccine authorization for pediatric patients. This study was performed at a single site, and findings may not be generalizable. However, as the study was conducted at the only pediatric hospital and academic pediatric endocrinology center in the state, our findings likely reflect true disease activity in the region. Additional multicenter prospective studies are needed to characterize national pediatric GD trends and better understand the relationship between SARS-CoV-2 infection, stress, and autoimmune thyroid disease in youths.

Disclosure

The authors have no multiplicity of interest to disclose.

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